

DomPII: A Self-Learning System with AI

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Abstract. DomPII is an open source educational platform based on artificial intelligence, designed to empower self-taught learners in any field of knowledge. Combining cutting-edge technologies such as LLMs, RAG, knowledge graphs, and dynamic interface generation (via MCPs), the system offers a personalized, interactive, and multimodal learning journey. More than a study environment, DomPII functions as an extension of thought, allowing the user to navigate through their own records, content, and cognitive connections through intelligent searches, timelines, and interdisciplinary visualizations. With flexible learning modes (Expert or Polymath), intelligent gamification, and a real reward system, DomPII transforms the act of learning into a creative, free, and stimulating experience where the user not only absorbs knowledge but transforms it.

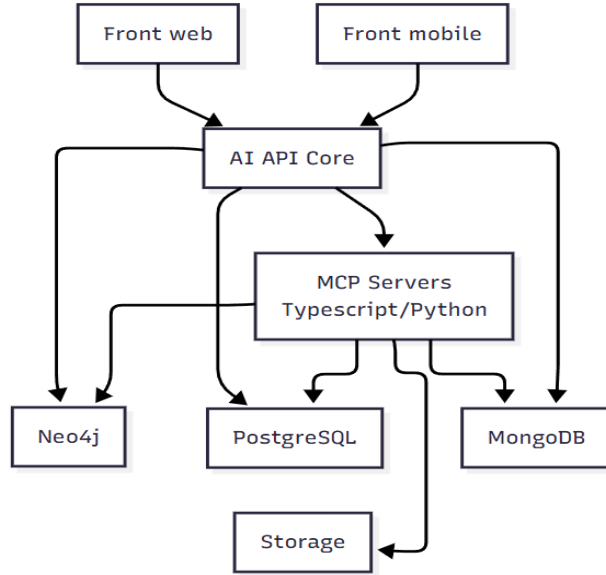
1. Introduction

DomPII is a self-learning platform based on artificial intelligence, created to offer a new way to explore, build, and apply knowledge. Instead of following rigid teaching paths, DomPII respects the uniqueness of each learner, providing flexibility, autonomy, and deep personalization of the educational process. At the heart of the platform is the innovative use of Large Language Models (LLMs), which act not only as intelligent tutors capable of explaining, questioning, and evaluating, but also as engines that generate complete learning interfaces and experiences.

Through a modular system called MCP (Model Context Protocol), the AI model itself dynamically builds the front-end of the application, generating interactive, multimodal, and responsive pages directly in HTML for canvas rendering, enabling unprecedented freedom in the design and structure of educational interactions.

DomPII transforms the learning environment into a living and expandable space, where the student can upload documents, videos, and audio that are processed and incorporated into their personal knowledge base. This base can be consulted naturally via RAG (Retrieval-Augmented Generation), relational graphs, or even an interactive timeline, allowing not only active study, but also continuous access to one's own intellectual history as if the system were a digital extension of the mind. In addition, DomPII integrates gamification mechanics, such as daily streak systems and real rewards (gift cards, configurable goals, and incentives from friends or family), promoting engagement without sacrificing cognitive depth. From the beginning, the user chooses between two modes: the Expert mode, focused on deepening a specific area, or the Polymath mode, which encourages transversal and interconnected exploration of multiple fields of knowledge.

2. System Architecture



2.1. Frontend

The DomPII frontend is intentionally minimalist, consisting only of a main chat, a settings menu, and a canvas area for HTML rendering. All the complexity of the interactive interfaces is delegated to the language model itself, which dynamically generates the visual and functional content through MCP tools. This approach significantly reduces the need for manual development of traditional interfaces, allowing AI to compose complete pages according to the context and user objectives. Additionally, it is also possible to use predefined layouts, which are dynamically filled by AI as needed. Details about this dynamic generation process are described in topic 2.3, which addresses how MCP works.

2.2. API

The DomPII API is based on REST architecture and offers endpoints for CRUD operations aimed at user customization and system configurations. It is integrated with the MCP protocol (Model Context Protocol), which acts as an intermediary between the system logic and language models, using AI as the main operation engine. The structure is compatible with different LLMs available in the market, including open models run locally, as long as they implement support for the MCP protocol. Much of the communication between frontend and backend occurs through streaming endpoints, connecting the user interface to MCP tools in a continuous, responsive, and multimodal way.

2.3. MCP Servers

The Model Context Protocol (MCP) is a standard created by Anthropic to connect language models to external tools through a unified communication interface, functioning as a "USB port" between the LLM and the execution environment. In DomPII, MCP is adopted as a fundamental layer that allows AI to operate directly on components of the system, whether they are media generators, HTML renderers, visual interfaces, or database handlers. Through MCP, it is possible to define expected input and output formats, as well as a brief description of the behavior of each tool. This allows creating MCP Tools that represent interactive interfaces or reusable functional blocks.

These tools can be predefined or generated on demand by AI, which understands their parameters and returns the complete representation of the content, typically in HTML ready to be rendered on the canvas. When an interface requires data manipulation (such as saving progress or recording user actions), a second MCP can be chained to the first, acting as its logical pair. The first generates the interface and action prompt; the second performs operations in the database according to the commands embedded in the context. Additionally, MCPs are used for generating physical and digital content, such as printable exercise PDFs, explainer videos, audio or dynamically created images to enrich the learning process.

Initially, these MCP tools need to be manually defined, but DomPII anticipates the development of two key features: An expandable MCP, capable of accepting from AI not only the data to be filled in, but also the interface description, and a generic MCP, designed for the registration and universal modification of data, which can be used for manipulating any type of data that the dynamic interface MCP may need.

With these features, it becomes possible to eliminate the need to develop new interfaces manually, delegating to AI the role of designer, implementer, and executor of visual and operational functionalities, as long as formal and security principles are respected. This architectural model transforms DomPII into a modular, flexible, and highly decoupled system. Each functionality is an autonomous MCP, which can be added, removed, or combined with others freely. This drastically reduces the complexity of maintenance and development, and at the same time makes room for a new type of educational system: interconnected, fluid, and perhaps even chaotic where screens don't follow a rigid hierarchy, but emerge as interactive environments that connect dynamically, according to each user's journey. This structure also enables a very high degree of customization, allowing each user, or even each community, to build their own DomPII with sets of MCP tools that meet their styles, objectives, and preferred themes, like a living educational system that adapts and reconfigures itself as it learns from those who use it. Examples of MCP:

Table 1: Examples of MCP tools in DomPII

MCP Type	Main Function	Examples in DomPII
Interface MCP	HTML interface generation and population	Quiz, Mind Map, Timeline
Data MCP	Database record manipulation	Save answers, Update progress
Media Generation MCP	Production of context-based digital content	Exercise PDFs, Review Images

Expanded Interface MCP	Dynamic creation of new interfaces via AI	On-demand screen generation
Generic Records MCP	Universal data registration and modification	Unpredictable records

Usage flow example:

- The user requests in the chat a "test about Artificial Intelligence".
- The API triggers the LLM, which identifies the need to use the "Quiz Generator" Interface MCP.
- The LLM accesses the quiz generation MCP and returns to the frontend the page filled with the requested theme.
- The user responds directly on the rendered interface, which communicates with the API in natural language.
- The answers are sent back to the LLM, which directs them to the "Quiz Result Recorder" Data MCP.
- The system registers the user's performance in the database.

2.4. Databases

DomPII uses different databases, combining their specific characteristics to offer the flexibility, scalability, and separation of responsibilities needed for the dynamic proposal of the system.

PostgreSQL, a relational database, is used to store more structured and stable information, such as user profiles, configurations, permissions, and other administrative records. It ensures referential integrity, performance in transactional queries, and security for the core data that support the basic operation of the platform.

MongoDB is used to store the raw knowledge generated and manipulated in the learning interfaces. Its document-oriented nature and lack of fixed schema allow each MCP to freely define its own data formats, maintaining independence between tools and adapting to the diversity of learning experiences.

Finally, Neo4j is used to represent relationships between content, contexts, and activities through intelligent graphs. These graphs are tuned with the support of AI algorithms, allowing both the system to build dynamic learning paths and the learner to explore their cognitive connections in personalized and visual ways. This combination of databases provides not only robustness and speed, but also opens paths to new forms of interaction with knowledge, which will be explored in more depth in the following topics.

2.5. Storage and RAG

DomPII incorporates a document and media storage system, allowing the user to upload content to complement their learning journey or even build formative paths exclusively from their own material.

All stored content, as well as learning records saved in MongoDB and their relationships in Neo4j, can be embedded through embedding techniques, generating a unique vector representation of each knowledge element. These vectors allow the system, using Retrieval-Augmented Generation (RAG), to retrieve relevant information efficiently and semantically, responding to user requests in a personalized and

contextualized way. This architecture transforms DomPII into a true extension of the learner's memory, a digital second brain where knowledge is not only stored, but can also be explored in innovative ways.

Access to this content can occur through different visual MCPs, which represent knowledge from various perspectives, such as mind maps, interactive chronological records, or other dynamic interfaces that can be continuously developed to enrich the experience of navigation and discovery.

3. Gamification

DomPII incorporates elements of gamification to strengthen engagement and continuity of users' learning journey. A daily streak system will be implemented, which counts the number of consecutive days in which the user interacts with the platform, encouraging the creation of consistent study habits. This dynamic can be explored in various ways, including integration with a reward system similar to a vault, where prizes are stored and unlocked upon reaching specific goals.

Users can add their own gift cards or receive contributions from friends and family, configuring personalized conditions for redemption. The goals for releasing rewards will be flexible and may include criteria such as the number of consecutive days of use, total time dedicated to the platform, progress in the studied content, among other metrics that may be incorporated as the system evolves. This mechanism aims to transform the learning process into a tangible experience of achievement, reinforcing the cycle of motivation and personal progress.

4. Limitations and Challenges

As the volume of information stored by the user grows within the system, it can become challenging for language models to handle extensive contexts efficiently. However, the continuous evolution of LLMs, especially the accelerated increase in token window size, combined with good practices in knowledge representation and compression, allows significantly mitigating this limitation.

Another challenge lies in the modular and active nature of DomPII. Users accustomed to traditional learning methods, which are more passive and linearly structured, may initially find the freedom and dynamics offered by the platform strange. However, the very modular architecture based on MCPs allows the creation of more directed flows and adaptive interfaces, offering more linear navigation and study options for those who prefer a more conventional learning experience. These factors, far from being definitive barriers, are considered opportunities for continuous evolution of the system, both in supporting diversity of learner profiles and in adopting the most advanced technologies in language processing.

5. Conclusion

DomPII represents a new approach to learning platforms: more open, flexible, deeply integrated with the evolution of artificial intelligence, and directed towards active and independent learning, without the need for traditional institutions.

By structuring its core around modularity, personalization, and learner autonomy, the system positions itself not only as an educational tool, but as an active extension of thought and knowledge construction. By adopting concepts such as the Model Context Protocol, the use of specialized databases, and RAG-assisted search, DomPII offers an architecture capable of growing organically with its users and keeping pace with AI advances continuously.

Although technical and pedagogical challenges are present, they are viewed as a natural part of the innovation process. The flexibility of the system ensures that it can adapt to new models, new user profiles, and new ways of exploring knowledge. With DomPII, learning is no longer just about accumulating information: it becomes a creative, dynamic, and personal act where each journey is unique, and where each achievement is also a real expansion of what we are capable of knowing and building.